Summer Research Internship Scheme - 2025

Internship report on

Fabrication and Analysis of wide band gap ZnO, Ga_2O_3 and SnO_2 UV Photodetectors

submitted to

Indian Institute of Technology (Indian School of Mines), Dhanbad

BY

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(ADMISSION No.: 21JE0522)

UNDER THE SUPERVISION OF

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INDIAN INSTITUTE OF TECHNOLOGY (INDIAN SCHOOL OF MINES), DHANBAD

July,2025

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DHANBAD – 826004 (JHARKHAND), INDIA

Date: July 9,2025

CERTIFICATE

Certified that the work presented in the dissertation entitled "Fabrication and Analysis of wide band gap ZnO, Ga_2O_3 and SnO_2 UV Photodetectors" is an authentic record of the research work carried out by Mageshan Kp under my supervision as part of Summer Research Internship Scheme 2025 at Indian Institute of Technology (ISM) Dhanbad during the period May2025 to July2025. Further that no part of this dissertation has been presented previously for the award of any other degree or internship.

Approved and forwarded by

Dr. R.Thangavel

(Supervisor)

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Declaration

I declare that the internship report entitled "Fabrication and Analysis of wide band gap ZnO, Ga_2O_3 and SnO_2 UV photodetectors" submitted by me for Summer Research Internship Scheme is the record of research work carried out by me under the guidance of Prof.R.Thangavel, Associate Professor, Department of Physics, IIT(ISM) Dhanbad and has not formed the basis for the award of any degree or internship in other university.

Mageshan Kp

Admission No: 21JE0522

Indian Institute of Technology (ISM) Dhanbad

Date: 9 July 2025

Abstract

This report presents the synthesis and characterization of UV-sensitive materials including SnO_2 quantum dots (QDs), ZnO nanorods, and Ga_2O_3 Nanorods for photodetector applications. SnO_2 QDs were synthesized using the Schlenk line method and characterized by XRD, FESEM, UV-Vis, and X'Pert HighScore software. ZnO nanorods were grown hydrothermally on silica and IDE substrates using 1.0 M and 0.5 M precursor concentrations, respectively, and analyzed using FESEM and I–V characteristics. Ga_2O_3 thin films were deposited via e-beam evaporation and characterized by XRD, UV-Vis, and I–V measurements. Each material showed unique photoresponse properties relevant to different UV detection regions.

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1. Introduction

Photodetectors sensitive to ultraviolet (UV) light are essential in applications ranging from flame sensing and environmental monitoring to bioimaging and secure communications. Wide bandgap metal oxides like SnO_2, ZnO, and Ga_2O_3 offer excellent thermal and chemical stability, low dark current, and strong UV absorption, making them suitable candidates. This work investigates each of these materials in different geometries and compares their structural and optoelectronic performance under UV illumination.

2. Experimental Methods

2.1 SnO_2 Quantum Dot Synthesis

SnO_2 QDs were synthesized via the Schlenk line method under inert atmosphere. Precursors were thermally decomposed under vacuum to yield uniform SnO_2 nanoparticles.

2.2 ZnO Nanorod Growth

ZnO nanorods were grown using a hydrothermal technique. A 1.0 M precursor solution was used for growth on plain silica substrates. For IDE substrates, a seed layer was first spin-coated and 0.5 M precursor solution was used for the growth of nanorods.

2.3 Ga_2O_3 Nanorod Growth

Ga_2O_3 was deposited on IDE substrates using e-beam evaporation growing Ga_2O_3 nanorods. The films were subsequently annealed to promote crystallinity and phase formation**.**

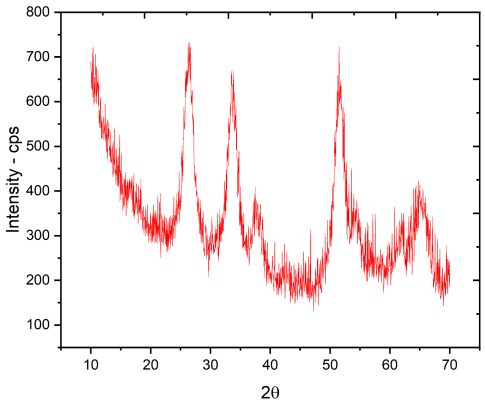
2.4 Characterization Techniques

- XRD: Used to identify crystalline phases (SnO_2 and Ga_2O_3)
- FESEM: Used to observe surface morphology of SnO_2 QDs and ZnO nanorods
- UV-Vis: Used to determine band gap of SnO_2 and Ga_2O_3
- I–V Measurements: Carried out on ZnO and Ga_2O_3 devices using gold IDE contacts, under dark and UV illumination

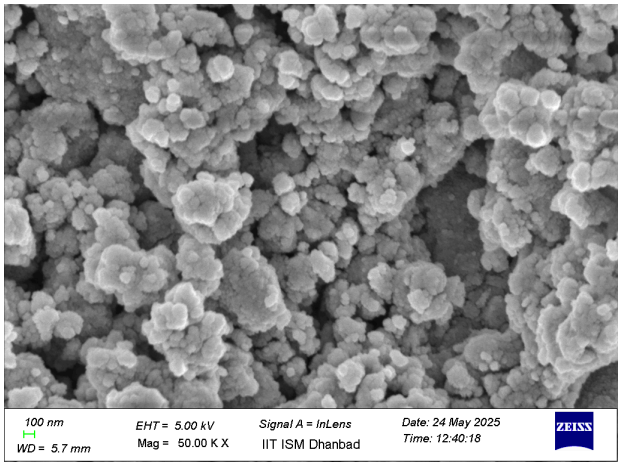
3. Results and Discussion

3.1 SnO_2 Quantum Dots

The XRD pattern of SnO_2 quantum dots confirmed the formation of the tetragonal rutile phase, with peaks matching standard JCPDS, indicating nanocrystalline behavior.

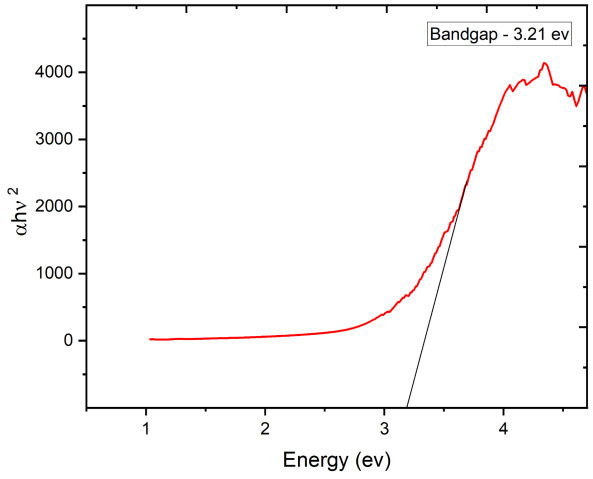


X'Pert HighScore analysis software was used to refine the XRD data and confirm peak indexing and phase purity. No secondary phases were detected.



FESEM images showed nearly spherical nanoparticles with uniform distribution, further supporting the quantum dot morphology. The small particle size contributes to quantum confinement effects.

The UV-Vis absorption spectrum displayed a strong absorption edge around 390 nm, indicating a wide optical bandgap. Tauc plot analysis estimated the bandgap to be approximately 3.21 eV, slightly lower than bulk SnO_2 (~3.6 eV), which is consistent with quantum confinement effects.

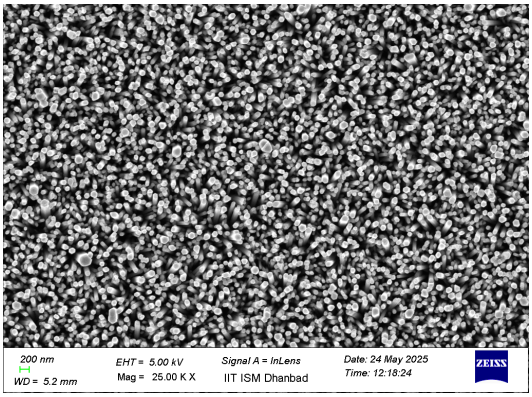


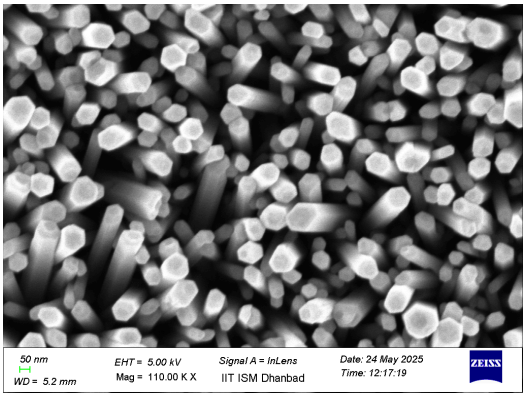
These features confirm the successful synthesis of SnO_2 quantum dots with enhanced optical properties, making them suitable for UV detection applications.

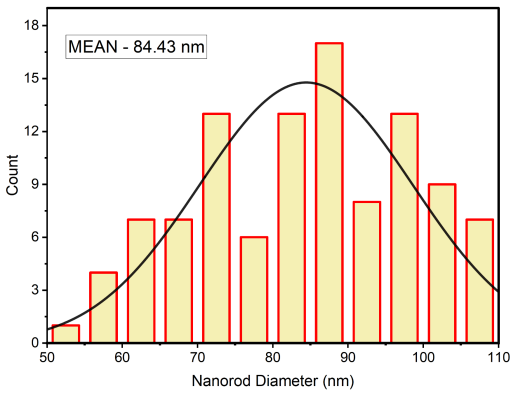
3.2 ZnO Nanorods

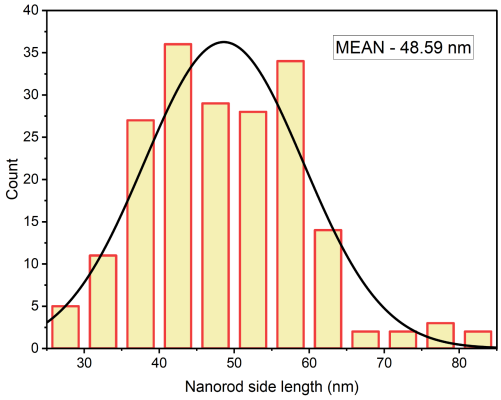
FESEM analysis revealed the effect of precursor concentration on the morphology of ZnO nanorods. At 1.0 M precursor concentration (on silica substrate), the nanorods appeared thicker and less densely packed, with an average diameter of ~98.37 nm. In contrast, when the precursor concentration was reduced to 0.5 M and grown on IDE substrate with a seed layer, the nanorods exhibited improved alignment and a narrower diameter of ~106.20 nm, indicating more controlled vertical growth.

1M precursor solution (Trial 1) :

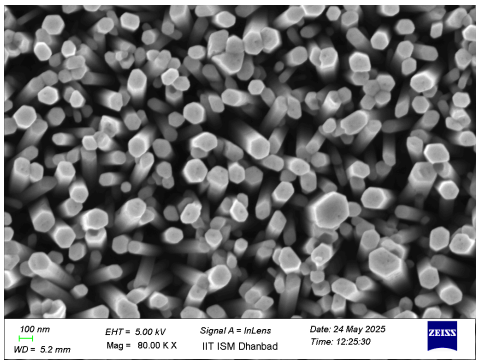


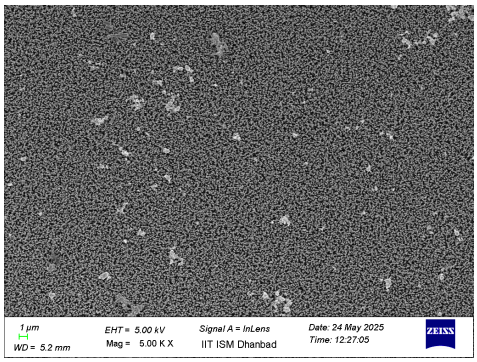


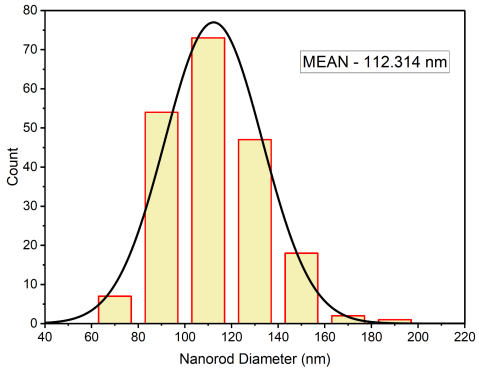


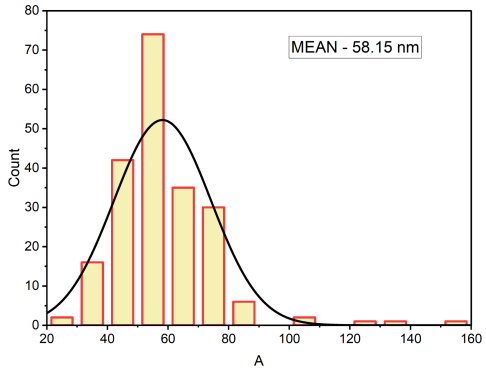


1M precursor solution (Trial 2) :





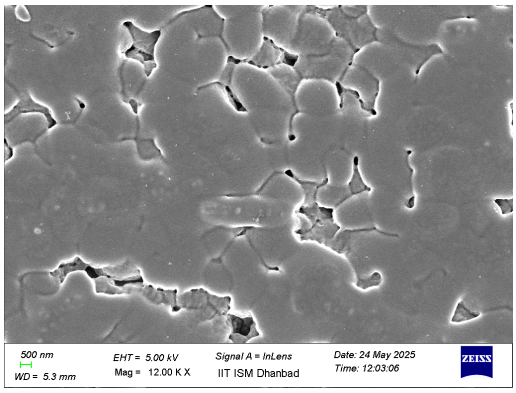


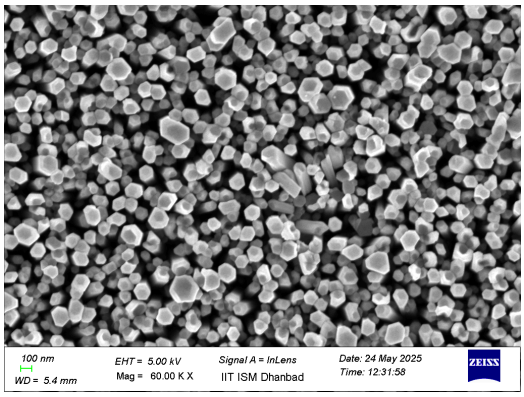


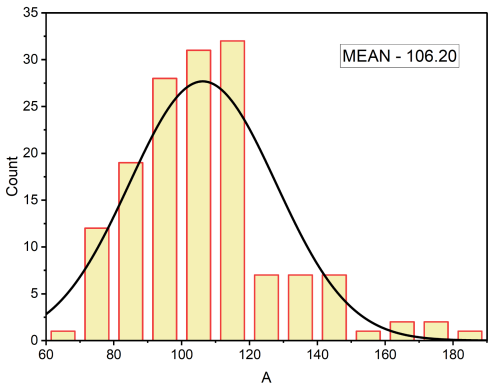
This change in aspect ratio and alignment has a direct influence on the electrical behavior of the photodetector. The IDE structure was selectively coated with ZnO only on the finger electrodes, leaving the contact pads exposed. This configuration promotes efficient charge separation and transport along the nanorods under UV illumination.

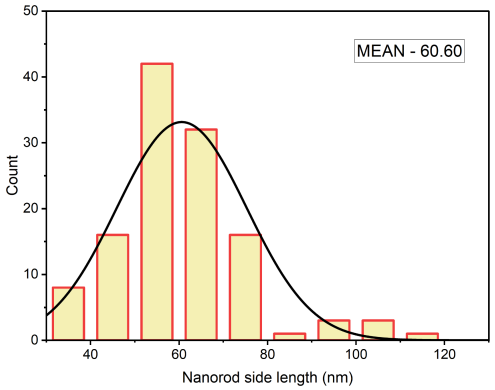
0.5 M precursor solution on IDE substrate:

Seed layer **Nanorods**

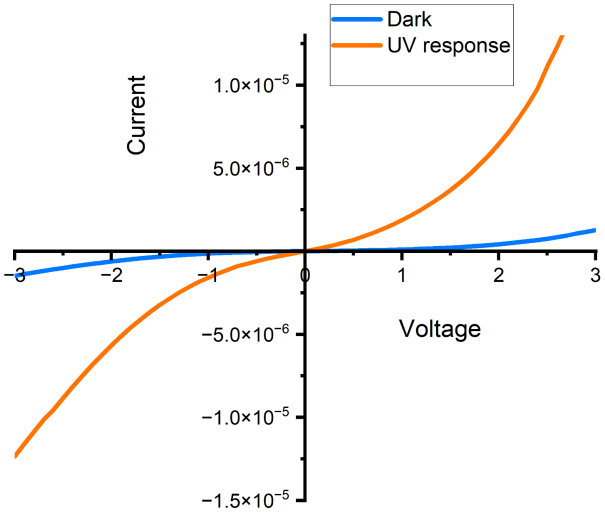








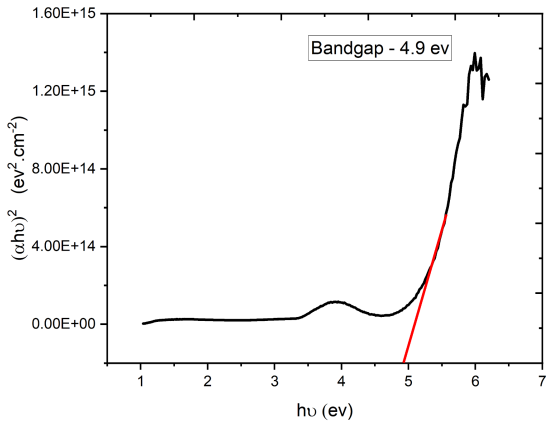
I–V measurements under dark and UV conditions showed a clear enhancement in photocurrent response. Under UV exposure, the forward and reverse currents increased symmetrically due to the photogeneration of electron–hole pairs in the ZnO nanorods. The I–V curve exhibited near-linear behavior, indicating ohmic-like contact between the ZnO and Au IDEs. The device showed a photo-to-dark current ratio greater than 10, confirming effective photoconductive UV detection.



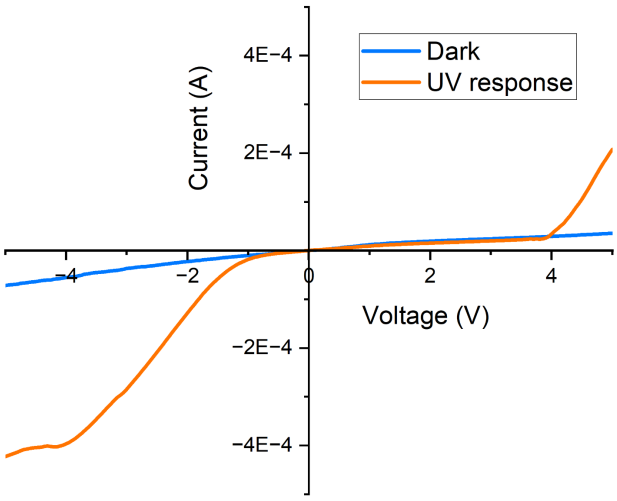
3.3 Ga_2O_3 Nanorods

XRD analysis confirmed the formation of β-Ga_2O_3 phase with strong and sharp diffraction peaks, indicating good crystallinity. No secondary phases were observed.

The UV-Vis absorption spectrum of the Ga_2O_3 nanorods showed a steep absorption edge near 260 nm. The optical bandgap, calculated from the Tauc plot, was approximately 4.9 eV, in agreement with reported values for β-Ga_2O_3, confirming its suitability for UVC detection.



The I–V characteristics under dark conditions revealed low current levels and near-linear behavior, indicating high resistivity and low leakage. Upon UV illumination, a significant increase in photocurrent was observed, especially under higher bias (>±4 V). The photoresponse was symmetric and non-linear, suggesting bulk photoconductivity rather than Schottky behavior. This is attributed to possible oxygen vacancies and high n-type conductivity in the e-beam grown Ga_2O_3 nanorod, resulting in near-ohmic Au contact behavior.



4. CONCLUSION

This report demonstrates the synthesis and characterization of three wide-bandgap semiconductors—SnO_2, ZnO, and Ga_2O_3—for UV photodetector applications. SnO_2 quantum dots synthesized via the Schlenk line method showed strong UV absorption and a widened bandgap due to quantum confinement. ZnO nanorods grown hydrothermally on different substrates revealed morphology-dependent photoresponse, with the IDE-based device showing efficient photoconduction and a high photo-to-dark current ratio. Ga_2O_3 nanorod growth by e-beam evaporation exhibited a wide bandgap and strong UVC sensitivity, with symmetric and nonlinear I–V behavior under illumination. The combined results highlight the potential of these oxide nanostructures for scalable and wavelength-selective UV photodetectors.